



Intestinal Protozoa — *Giardia*, *Cryptosporidium*, *Balantidium*

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

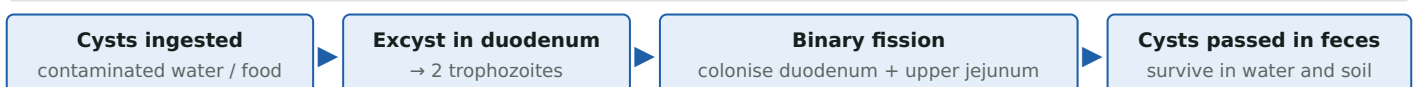
1. Which protozoan is the only one found in the lumen of the human small intestine, and what does its trophozoite look like?
2. Which stain is the method of choice for *Cryptosporidium* oocysts, and how do they look?
3. Which intestinal protozoan is a ciliate acquired from pigs, and what is the drug of choice?

*Guessing wrong before reading still improves retention — attempt all three, then check as you go.***1 Three protozoa — one discriminator grid**

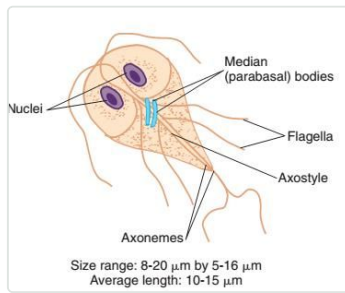
	<i>Giardia lamblia</i>	<i>Cryptosporidium parvum</i>	<i>Balantidium coli</i>
Type	Flagellate	Obligate intracellular apicomplexan	Ciliate — the only ciliate infecting humans; largest human protozoan
Infective form	Mature cyst (4 nuclei), ~12 × 8 µm	Sporulated oocyst , ~5 µm	Cyst , 40–60 µm
Site	Duodenum and upper jejunum (lumen)	Small intestine (also stomach, appendix, colon, rectum, lungs)	Large intestine
Picture	Foul-smelling, fatty stools, flatulence, cramps; no blood usually	Self-limiting watery diarrhea; profuse, persistent in AIDS	Dysentery like amoebiasis; ulcers, submucosal abscesses
Best diagnostic clue	Pear-shaped trophozoite, sucking disc , "falling leaf" motility	Modified acid-fast red spheres; not stained by iodine	Huge, rapidly motile trophozoite with kidney-shaped macronucleus
Reservoir / source	Contaminated water, food; person-to-person	Water-borne; humans, birds, mammals (zoonotic)	Zoonosis — pigs, monkeys, rats; human carriers
Treatment	Metronidazole or tinidazole	Nitazoxanide + supportive care	Tetracycline

2 *Giardia lamblia* — script

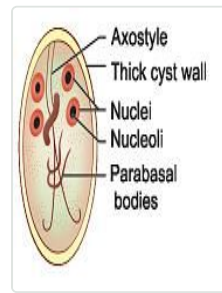
WHO / WHERE	One of the most common protozoan infections; poor sanitation , tropical / subtropical; traveller's diarrhoea via contaminated water; common in children; person-to-person spread in children, male homosexuals, mentally ill persons
WHY (biology)	Trophozoite ~15 × 9 × 4 µm: pear / tennis-racket shaped, convex dorsum, concave ventral sucking disc , bilaterally symmetrical, 2 nuclei , 4 pairs of flagella , paired axostyles, two median bodies; slow "falling leaf" motility . Cyst : oval ~12 × 8 µm, hyaline wall, 2 nuclei when immature → 4 nuclei when mature; survives months in water
HOW (pathogenesis)	No tissue invasion ; attaches by the sucking disc → villous damage, epithelial apoptosis, lymphocytic infiltration; variant-specific surface proteins (VSPs) aid virulence; infective dose 10–100 cysts
WHAT you see	Often asymptomatic; symptoms after 1–2 weeks . Acute: foul diarrhea, flatulence, cramps, bloating. Chronic: steatorrhea , malabsorption , weight loss, fatigue; stool with mucus and fat, usually no blood ; children: chronic diarrhea and nutritional deficiency; rare uveitis / retinal hemorrhage



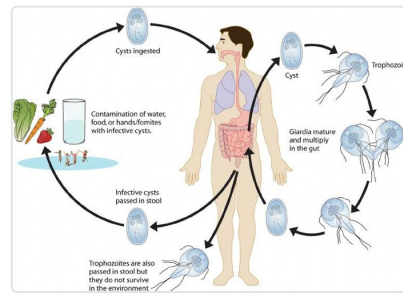
Life cycle in ONE host; infective form = mature cyst.



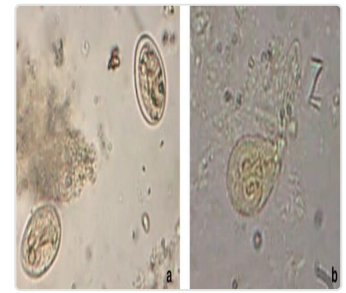
Trophozoite.



Cyst.



Life cycle.



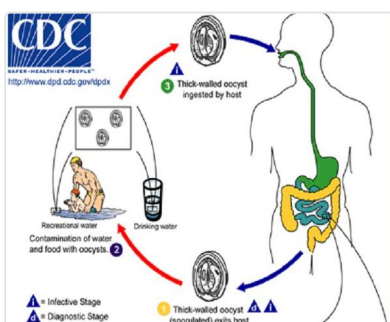
Cysts on wet mount.

Diagnosis	Detail
Stool	Pale, foul, fatty, floating stools; cysts / trophozoites on saline or iodine wet mount
Antigen / antibody / PCR	Stool antigen ELISA / IFT = active infection ; antibody tests mainly epidemiological; PCR for DNA
Duodenal contents	String test (Entero-test) : swallowed gelatin capsule with a thread collects duodenal fluid → trophozoites
Treatment / prevention	Detail
Drugs	Metronidazole or tinidazole first line; furazolidone or nitazoxanide (esp. children); paromomycin in pregnancy
Prevention	Hygiene; avoid food / water contamination; community chlorination does NOT inactivate cysts — boil or filter water

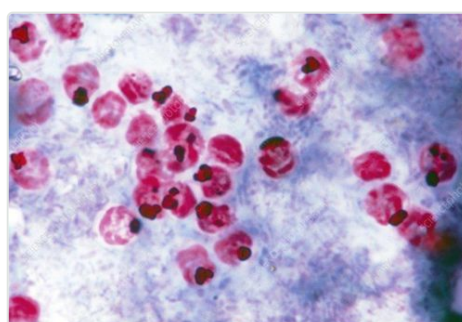
3 *Cryptosporidium* — script

WHO	Worldwide; mainly <i>C. hominis</i> and <i>C. parvum</i> ; fecal-oral , person-to-person; outbreaks in day-care centres and hospitals ; risk: AIDS / immunodeficiency , young children, older adults, handling infected cattle, swallowing recreational water; "most common water-borne illness in the world"
WHY (biology)	Oocyst ~5 µm, spherical/oval, acid-fast , does NOT stain with iodine , 4 crescent-shaped sporozoites ; thick-walled = environmentally resistant; thin-walled (~20%) = autoinfection ; highly resistant to disinfectants
HOW (cycle)	Sexual + asexual phases in a single host (humans, birds, mammals); infective form = sporulated oocyst
WHAT you see	Immunocompetent: asymptomatic or self-limiting watery diarrhea , pain, nausea, weight loss. AIDS: chronic, persistent, remarkably profuse diarrhea , dehydration, electrolyte loss, marked weight loss; rarely respiratory disease

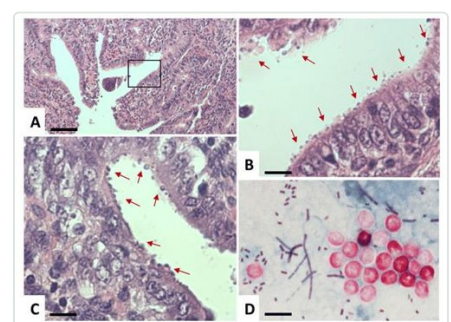
Diagnosis / treatment	Detail
Method of choice	Modified acid-fast stain of stool — oocysts appear as red spheres
Others	Immunofluorescence (antibody); stool antigen ELISA; PCR; small-bowel biopsy (parasitophorous vacuoles on the epithelial surface)
Treatment	Self-limiting in most immunocompetent patients; no effective specific therapy ; nitazoxanide + rehydration, electrolytes, nutrients
Prevention	Personal hygiene; avoid food / water contamination with human and animal feces; proper water treatment



Life cycle (CDC).



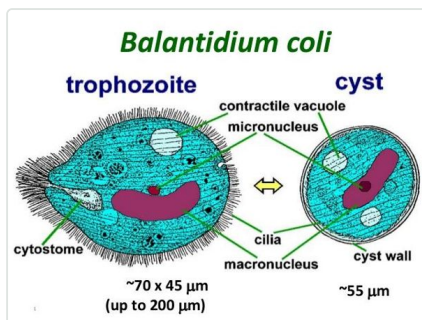
Modified acid-fast: red oocysts.



Biopsy: organisms on the epithelial surface.

4 *Balantidium coli* — script

WHO / WHERE	Phylum Ciliophora ; worldwide but uncommon ; highest in New Guinea (close human–pig contact); lives in the large intestine of humans, pigs, monkeys
WHY (biology)	Trophozoite : 60–70 μm , ovoid, kidney-shaped macronucleus + small micronucleus, food and contractile vacuoles; motile, invasive ; found in dysenteric stool. Cyst : spherical, 40–60 μm , thick double-layered wall ; infective ; common in chronic cases and carriers
HOW (transmission)	Zoonosis : ingestion of food / water contaminated with cysts from pigs, other animals or human carriers ; one host only (monoxenous); human = accidental host; reservoirs: pig, monkey, rat
WHAT you see	Usually commensal / asymptomatic ; disease with malnutrition, alcoholism, reduced host resistance . Trophozoites invade colonic mucosa → ulcers and submucosal abscesses ; diarrhea, dysentery, cramps, nausea, vomiting — resembles amoebiasis, but extraintestinal invasion is uncommon
Diagnosis / Rx	Trophozoites in diarrheal stool , cysts in formed stool (large size, macronucleus, rapid motility); biopsy if stool negative. Tetracycline is the drug of choice; alternatively doxycycline ; metronidazole in some cases
Prevention	Avoid feces contamination of food / water; prevent human-pig contact ; better sanitation; treat cyst shedders



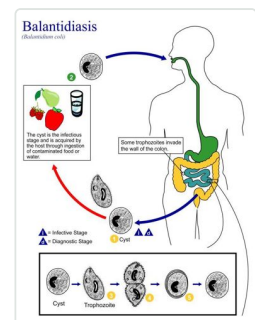
Trophozoite and cyst.



Trophozoite (stained).



Cyst.



Life cycle (CDC).

P Must-remember pearls

- **Giardia** = only protozoan in the **lumen** of the small intestine; **pear-shaped, sucking disc, 4 pairs of flagella**; infective **cyst (4 nuclei)**; fatty foul stool, no blood; **metronidazole / tinidazole**.
- **Chlorination does not inactivate Giardia cysts** — boil or filter; **string test** samples duodenal trophozoites.
- **Cryptosporidium** = **acid-fast, iodine-negative 5 μm oocysts**; **AIDS** → **profuse persistent diarrhea**; no effective specific therapy — **nitazoxanide + rehydration**.
- **Balantidium** = only **ciliate, largest** protozoan; **pig zoonosis**; dysentery like amoebiasis but **rarely extraintestinal**; **tetracycline**.
- Stool prep tells them apart: cysts + trophozoites (Giardia), **red acid-fast spheres** (Crypto), **giant ciliated trophozoite** (Balantidium).

Q Test yourself

1. A traveller has foul-smelling, fatty, floating stools without blood. A wet mount shows pear-shaped trophozoites with two nuclei and "falling leaf" motility. Which drug is first-line?

- A. Metronidazole
B. Tetracycline
C. Praziquantel
D. Mebendazole

2. A patient with AIDS has profuse chronic watery diarrhea; stool stained by modified acid-fast shows 5 μm red spheres. Diagnosis?

- A. *Giardia lamblia*
B. *Cryptosporidium*
C. *Balantidium coli*
D. *Enterobius vermicularis*

3. A pig farmer with malnutrition has dysentery; stool shows a very large ciliated trophozoite with a kidney-shaped macronucleus. Which drug is preferred?

- | | |
|-----------------|----------------------|
| A. Tetracycline | B. Nitazoxanide only |
| C. Paromomycin | D. Praziquantel |

4. Campers drank lake water treated with chlorine and later developed giardiasis. What should have been done?

- | | |
|-----------------------------|----------------------|
| A. Boil or filter the water | B. Add more chlorine |
| C. Store water for 24 h | D. Use antacids |

5. Stool microscopy is negative in a patient strongly suspected of giardiasis. Which test samples duodenal contents?

- | | |
|------------------------------|-----------------------------|
| A. String test (Entero-test) | B. Perianal tape |
| C. Kato-Katz | D. Modified acid-fast stain |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A — Giardiasis: metronidazole or tinidazole first line.

2 — B — Modified acid-fast red spheres are Cryptosporidium oocysts; AIDS gives severe persistent diarrhea.

3 — A — Balantidium coli: tetracycline is the drug of choice (doxycycline alternative).

4 — A — Chlorination does not inactivate Giardia cysts; boiling and filtration are needed.

5 — A — A swallowed gelatin capsule on a thread collects duodenal fluid for trophozoites.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Microbiology GIT Block · MEDucation by Mattin Majid · GIT SGL 5 slides, 2026-2027
Study alongside the lecture, not instead of it.